

Question 1**[10]**

A basketball player grabbing a rebound jumps 76.0 cm vertically. How much total time (ascent and descent) does the player spend

- (a) in the top 15.0 cm of this jump? [4]
 (b) in the bottom 15.0 cm? (The player seems to hang in the air at the top.) [6]

Question 2**[10]**

A proton initially has $\vec{v} = 4\hat{i} - 2\hat{j} + 3\hat{k}$ and then 4.0 seconds later has $\vec{v} = -2\hat{i} - 2\hat{j} + 5\hat{k}$ (in meters per second). For that 4.0 seconds, what are:

- (a) the proton's average acceleration $\vec{a}_{avg}$ in unit vector notation, [4]
 (b) the magnitude of $\vec{a}_{avg}$, [3]
 (c) the angle between $\vec{a}_{avg}$ and the positive direction of the x axis? [3]

Question 3**[12]**

- (a) Show that the trajectory (path) of a particle in projectile motion is given by

$$y = (\tan\theta_0)x - \frac{gx^2}{2(v_0\cos\theta_0)^2}. \quad [6]$$

 (b) A soccer ball is kicked from the ground with an initial speed of 19.5 m/s at an upward angle of 45° . A player 55 m away in the direction of the kick starts running to meet the ball at that instant. What must be his average speed if he is to meet the ball just before it hits the ground? [6]

Question 4**[6]**

Figure 1 shows a cord attached to a cart that can slide along a frictionless horizontal rail aligned along an x axis. The left end of the cord is pulled over a pulley, of negligible mass and negligible friction, and at cord height $h = 1.20$ m, so the cart slides from $x_1 = 3.00$ m to $x_2 = 1.00$ m. During the move, the tension in the cord is a constant 25.0 N. What is the change in the kinetic energy of the cart during the move?

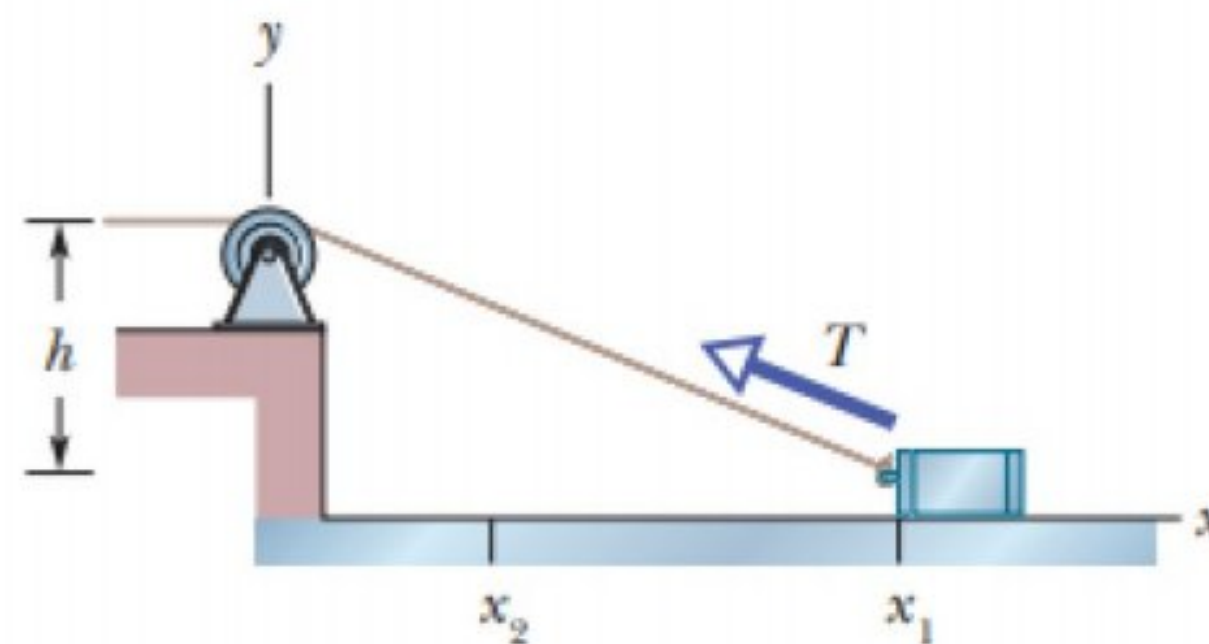


Figure 1: Question 4

Question 5**[8]**

A banked circular highway curve is designed for traffic moving at 60 km/h. The radius of the curve is 200 m. Traffic is moving along the highway at 40 km/h on a rainy day. What is the minimum coefficient of friction between tires and road that will allow cars to take the turn without sliding off the road? (Assume the cars do not have negative lift.)

Question 6**[3]**

Explain what is meant by elastic and inelastic collisions.

Question 7**[6]**

In Figure 2(a), a 3.50 g bullet is fired horizontally at two blocks at rest on a frictionless table. The bullet passes through block 1 (mass 1.20 kg) and embeds itself in block 2 (mass 1.80 kg). The block ends up with speeds $v_1 = 0.630$ m/s and $v_2 = 1.40$ m/s in Figure 2(b). Neglecting the material removed from block 1 by the bullet, find the speed of the bullet as it:

- (a) leaves block 1,
- (b) enters block 1.

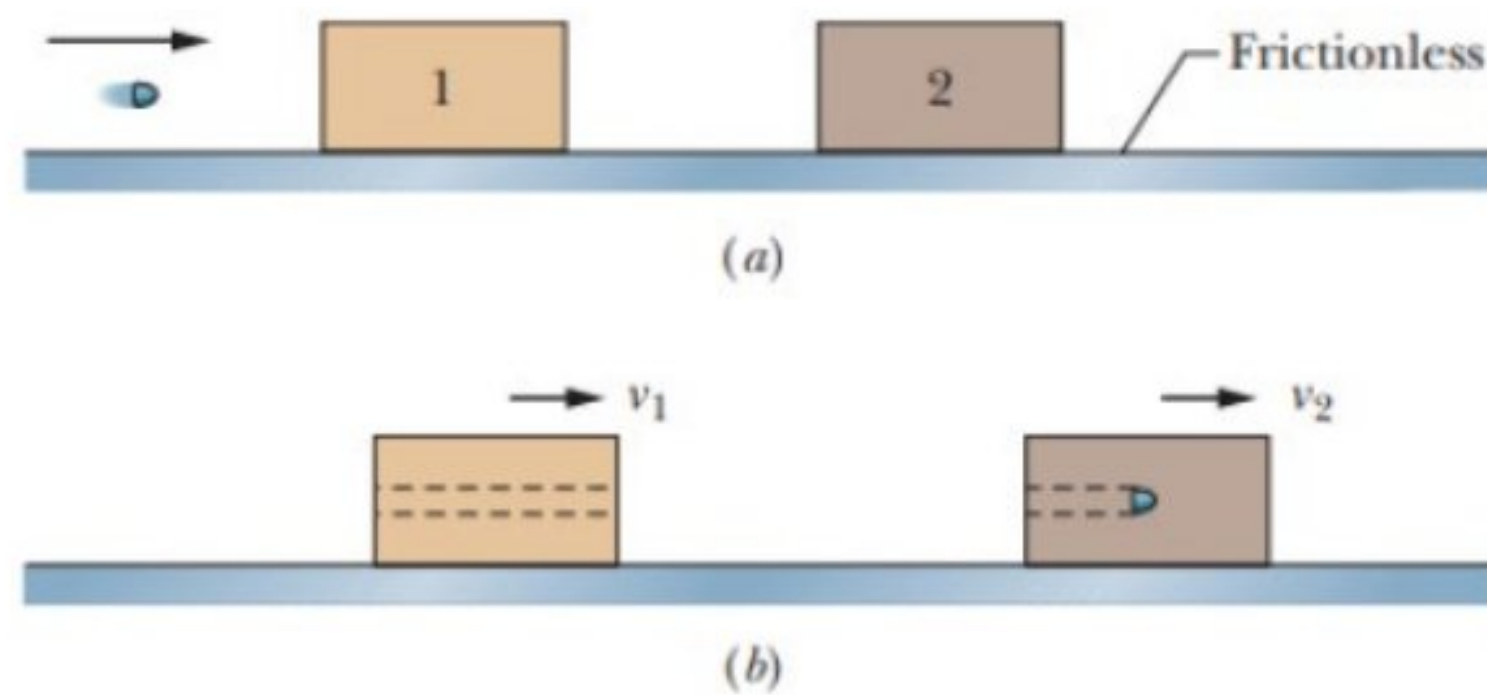
[3]**[3]**

Figure 2: Question 7

Question 8**[6]**

A man stands on a platform that is rotating (without friction) with an angular speed of 1.2 rev/s; his arms are outstretched, and he holds a brick in each hand. The rotational inertia of the system consisting of the man, bricks, and platform about the central vertical axis of the platform is $6.0 \text{ kg}\cdot\text{m}^2$. If by moving the bricks the man decreases the rotational inertia of the system to $2.0 \text{ kg}\cdot\text{m}^2$, what are:

- a) the resulting angular speed of the platform?
- b) the ratio of the new kinetic energy of the system to the original kinetic energy?
- c) What source provided the added kinetic energy?

[2]**[2]****[2]****Question 9****[4]**

In Figure 3, trying to get his car out of the mud, a man ties one end of the rope around the front bumper and the other end tightly around a utility pole 18 m away. He then pushes sideways on the rope at its midpoint with a force of 550 N, displacing the center of the rope 0.30 m, but the car barely moves. What is the magnitude of the force on the car from the rope? (The rope stretches somewhat.)

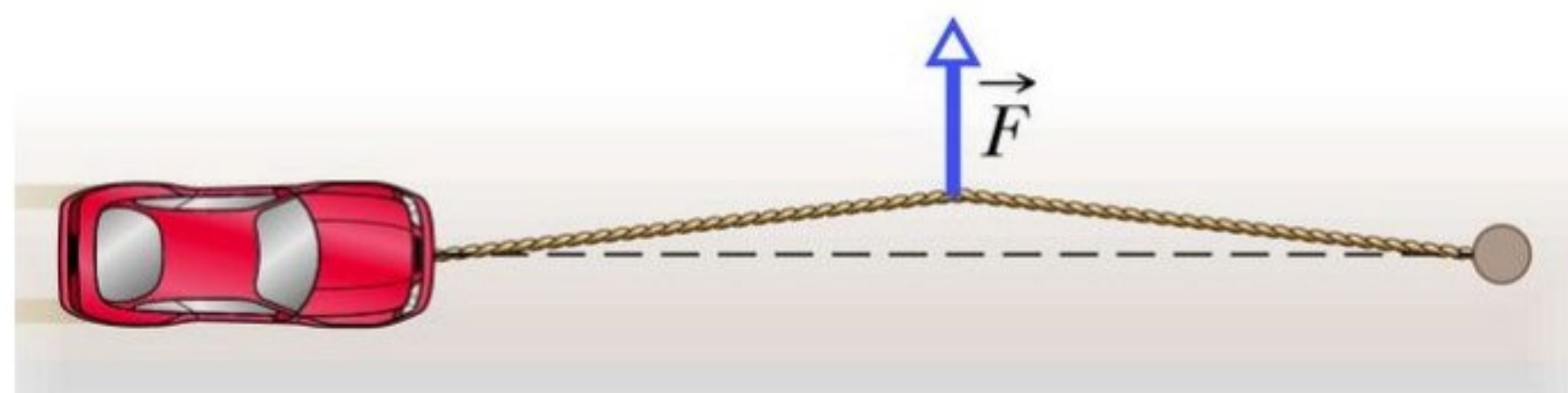


Figure 3: Question 9

Question 10

[5]

The water tank in Figure 4 has a hole in it close to the bottom of the tank.

- (a) Only using symbols, what is the speed v of the water exiting the water tank? [4]
- (b) If the height difference between the top of the water level inside the tank and the hole is 10 m, calculate the speed of the water exiting the water tank. [1]

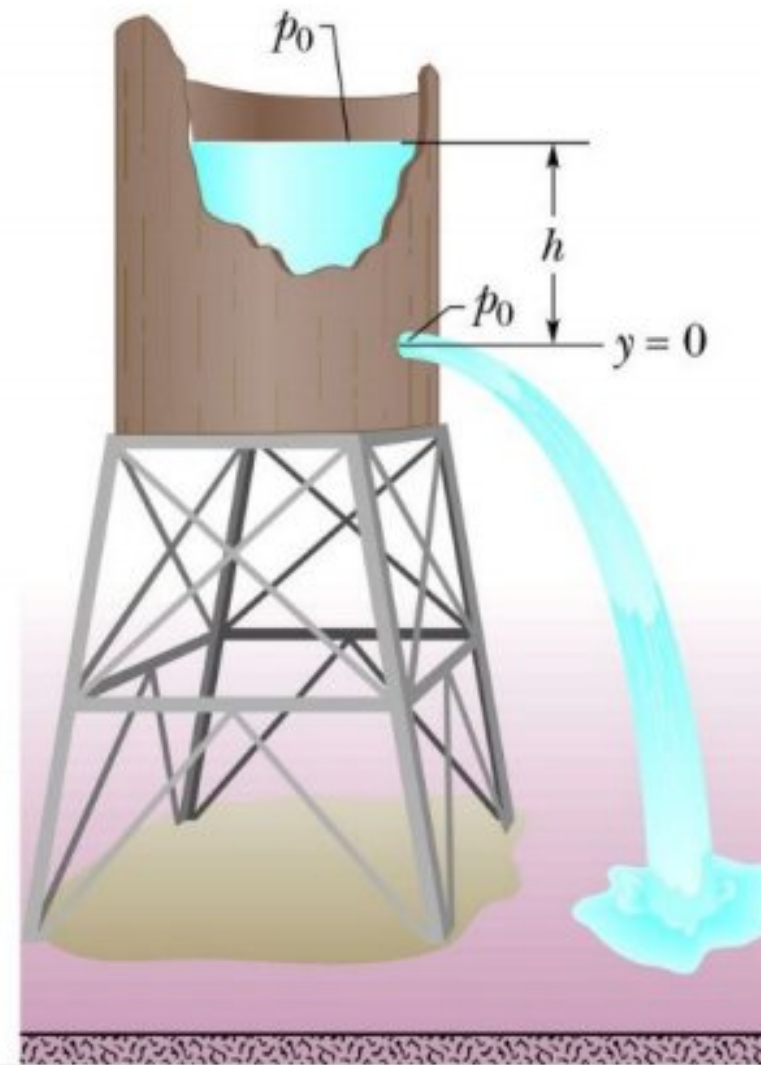


Figure 4: Question 10

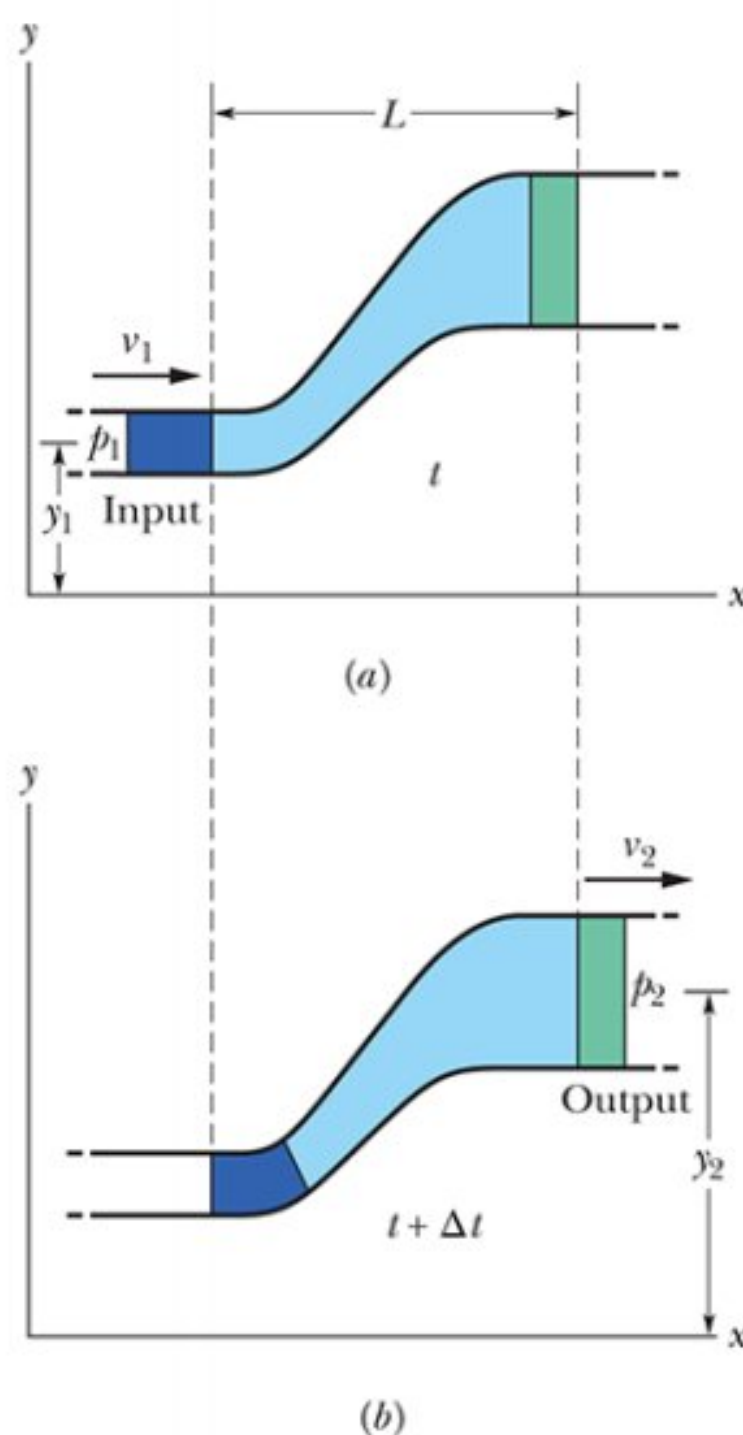
Question 11

[6]

Derive an equation for the pressure in a fluid at a specific point as a function of depth.

Question 12

[10]



Using the information and scenario presented in Figure 5, prove that Bernoulli's equation can be written as:

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2.$$

Figure 5: Fluid flows at a steady rate through a length L of a tube, from the input end at the left to the output end at the right. From time t in (a) to time $t + \Delta t$ in (b), the amount of fluid shown in dark blue enters the input end and the equal amount shown in green emerges from the output end.

Question 13**[9]**

A wave travelling along a string is described by: $y(x, t) = (0.0351\text{m})\sin [(52.3 \text{ rad/s})x + (2.52 \text{ rad/s})t]$ with x in meters and t in seconds.

- a) What is the wavelength of the wave? [3]
- b) What is the period of oscillation? [3]
- c) What is the frequency of the wave? [3]

Question 14**[4]**

A stationary motion detector sends sound waves of frequency 0.150 MHz toward a truck approaching at a speed of 45.0 m/s. What is the frequency of the waves reflected back to the detector?

Question 15**[4]**

A Carnot engine operates between 235°C and 115°C, absorbing 6.30×10^4 J per cycle at the higher temperature.

- a) What is the efficiency of the engine? [2]
- b) How much work per cycle is this engine capable of performing? [2]

Question 16**[3]**

One mole of oxygen (assume it to be an ideal gas) expands at a constant temperature T of 310 K from an initial volume V_i of 12 L to a final volume V_f of 19 L. How much work is done by the gas during the expansion?

Question 17**[4]**

Assume a heat of fusion $L_F = 333$ kJ/kg and a heat of vaporization $L_V = 2256$ kJ/kg for water.

- (a) What is the entropy change of a 12.0 g ice cube that melts completely in a bucket of water whose temperature is just above the freezing point of water? [2]
- (b) What is the entropy change of a 5.00 g spoonful of water that evaporates completely on a hot plate whose temperature is slightly above the boiling point of water? [2]

TOTAL: 110 marks